Imperial College London

AERO96005 - GROUP DESIGN PROJECT

Wholly Reusable Launch System (Recovery)

Subsystem layout of a Novel Synchropter Concept for the Aerial-Recovery of Falling Rocket Stages

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Abstract

This report details the selection process of the recovery capture mechanism, which is then explained further in the Capture report [1]. Avionics components were selected based on the mission profile and an avionics stack was suggested. Architecture of the system was designed bearing in mind that a later stage of development will involve upgrading to a fully autonomous platform, therefore the avionics was designed to be flexible and easy to modify. The placement of cameras around the UAV was made to provide the remote pilot full situational awareness. An electro-hydrostatic actuation system was then selected and sized to provide full control to the rotors and stabilator. After on board power requirements had been assessed, a generator was sized and backup battery selected.

WRR Team Members

Table 1: *Team Members and Roles.*

Name	Team Name	Team Code
Alessandro Bullitta Yassine Ghenima Cyril Gliner	Project co-ordination	WRR01
Chi Wing Cheng Alessandro Rossi Polvara Tin Hang Un Le Yin	Aerodynamics	WRR02
Anton Le Tejasva Malhotra **Atri Sharma** Alastair Wood	Structures & CAD	WRR03
Omar Arangath Thomas Cross Zuzanna Janusz **Sam Jenner** Jo Tan Tom Walker	Systems & Propulsion	WRR04
Filip Bastien Ananya Dubey Kai San Hon Andrew Ng India Tavares	Flight Mechanics, Performance & Control	WRR05
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†Sub-team leads are shown in **bold**

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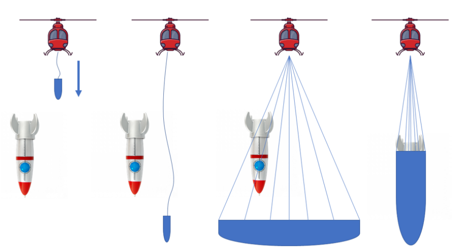
1 Introduction

The first phase of this project constituted a pre-conceptual design phase where 3 concept vehicles (fixed wing aircraft, helicopter and VTOL) were considered for the mission of capturing the falling stages of the WRL team Rocket. The helicopter was clearly favourable over the fixed wing due to it's higher manoeuvrability and opportunity for multiple capture attempts. After the launch team guaranteed a 50km landing area, the high range advantage of a VTOL became redundant and the project moved forward with a helicopter design.

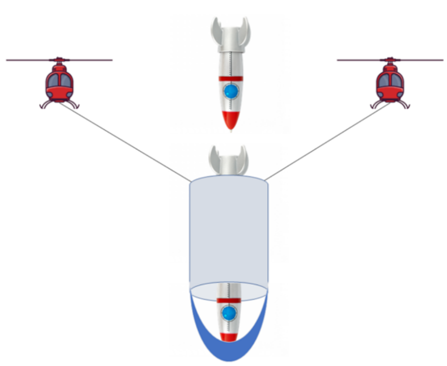
2 Recovery mechanism selection

After the decision was made that a helicopter would be the vehicle of choice, multiple concepts of recovery were considered. When selecting the final choice, the Technological Readiness Level (TRL) was strongly considered as this helped assess the feasibility of concept. Table 2 presents the strengths and weaknesses of considered concepts and Figure 1 display the action of each. Ultimately the Hook and Parafoil method was selected for it's high historic success rate [6] and high TRL. A detailed design of the recovery mechanism can be found in the related report [1].

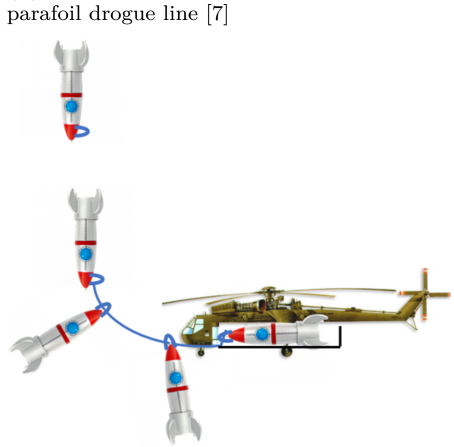


(a) Net is launched into trajectory of payload. Net expands via pyrotechnics and catches payload

(b) Helicopter capture line intercepts payload parafoil drogue line [7]



(c) Payload is decelerated when travelling through magnet and delivered into net suspended by team of helicopters



(d) Payload is caught on guide rail and swings into helicopter bay

Figure 1: Operation of cannon net, hook and parafoil, magnetic decelerator and guide ring recovery mechanisms

Table 2: *Summary of potential capture mechanisms*

Capture concept	Advantages	Disadvantages	TRL
Cannon-net	Capable of multiple attempts Cannon can be aimed at payload from any position	High jerk on helicopter Risk of line tangling	2
Hook and parafoil	Well established method Large capture window	Pilot precision required	8
Magnetic decelerator	Payload brought to near rest	High magnet mass Multiple helicopters required Payload must be magnetic	1
Guided ring	Simplest manufacture	Small capture window Large loading on helicopter High pilot precision required	3

3 Avionics

3.1 Architecture

The architecture of the avionics will determine how data is distributed, processed and outputted to sub-systems. Due to the fast development cycle of UAV technology and the expected future development of this helicopter, specifically augmenting it into a fully autonomous vehicle, it was chosen to move forward with an open architecture. In this model, the whole system avionics will be the aggregate of interconnected individual modules, which can be broken down into easier identified functional groups. Similarly, this grouping method will allow for system errors to be identified by module, this will allow for a more efficient diagnostic process. With this set-up, modules with similar data requirements can be grouped together, allowing for a higher computing performance, reduced cabling complexity and lower power requirements. Computationally expensive procedures will be transferred to redundant processing units, that will ease the workload of the main flight computer and other critical components. An NVIDA Jetson TX2 [8] has been chosen for this role for it's high processing power, but more importantly for it's adaptability to neural network processing, a function to be implemented as the system is shifted to a fully autonomous platform.
A shortlist of data bus standards was produced, their advantages and disadvantages are shown in Table 3. Spacewire was discarded as it is largely unsupported outside of the European Space Agency, therefore would require minor adaption. ARINC - 664 was chosen for it's universal use and strong specifications.

Table 3: *Summary of Databus standards*

Databus	Advantages	Disadvantages
MIL-1773	Low complexity High reliability	Lower bandwidth
ARINC - 664	Consistent data rate at high speeds Commercial off the shelf components	Lower data security
Spacewire IEEE - 1355-2	Low failure rate Auto detection of cable faults	Narrow user base

3.2 Avionics Sensor Health Assessment

Even with comprehensive testing, verification and validation of avionics sensing units and onboard software, heisenbugs and other rarely occurring events can manifest from the concurrency of multithreaded hardware. This can compromise the system reliability. In order to monitor the health of individual components, highlight anomalies and effectively isolate faulty components before real world problems arise, a Data Distribution Service is to be introduced. Data from the navigational, electric and inertial sensing units will be shared in real time to a third party service provider who will compare this data against their own trained algorithm to identify potential risks in the system and send back reports on the system health and reliability. 'Real Time Innovations' has been chosen as the service provider due to their previous work on US defence projects and in the aerospace industry [9].

3.3 Automatic flight control system

This mission to be completed requires moments of high precision, namely during the capture manoeuvre and landing with the payload. This, alongside the pilots reduced 'feel' of the aircraft due to being in a remote cockpit calls for a reduction of workload which is to be achieved with the use of an automatic flight control computer. The AFCS used is to be an integrated autopilot and Stability Augmentation System, of which the latter will always be active. The SAS is to have authority over the collective and cyclic. The is to be paired with a throttle governor system which will adjust throttle appropriately to achieve a constant RPM of the rotors. The combined system is to actively dampen any pitch and roll oscillations.
A suitable reference model of the flight computer is the Genesys IFR HeliSAS , which is in use on Sikorsky UH-60 [10]. This model was picked for it's uses in other heavier-lifting helicopters but also for the range of functions, including constant airspeed control and GPS course tracker. The most crucial use of the HeliSAS will be the constant glide slope function which will allow the pilot to carefully move into echelon position when executing the capture manoeuvre [1]. The AFCS is to be fed data from the altimeter, inertial sensing units, GPS, lidar and pressure sensors in order to most accurately track the current helicopter state.

3.4 Radar altimeter

A Radar altimeter is necessary, not only to conform with IFR UAS requirements but more importantly to provide the remote pilot a precise altitude when approaching the payload. Information from the Altimeter, supplemented by LIDAR will be of high importance during the payload landing phase of the mission. The Honeywell KRA 405B was selected due to it's compactness and 25,000ft range [11]. For reference, the mission is expected to start at maximum 20,000ft.

3.5 Transponder

Although during the mission an empty airspace is expected, convertibility for other mission profiles is still considered. Due to this, a Garmin GTX 335R Transponder will be equipped , facilitating Mode-S ADS-B communication with other aircraft and Aircraft Traffic Control to ensure safe flight trajectories [12]. This feature can also be used for additional position tracking data against Aircraft Traffic Control Towers.

3.6 Inertial navigation system/ Global positioning system

An INS/GPS has been chosen to provide highly accurate system state identification and positioning to the UAV, more specifically to the flight computer in order to provide full-state feedback and stability to the UAV. The Kearfott KN-4072A was selected for it's comprehensive state information, providing heading, attitude, velocity and position via a combination of 3-axis Monolithic ring laser gyros, fiber optic gyros and accelerometers. Pitch, roll, yaw and their respective rates are all estimated to within 0.5 degrees of their true value [13].
Where GPS data alone is insufficient for precisely locating the aircraft, Real Time Kinematic (RTK) positioning is to be used. This method compares the phase of the signal carrier wave from the UHF band against an interpolation of ground stations [14].
This neglects the content of the signal and uses information of the signal wavelength and number of cycles in the signal to calculate the position. The error of this position is the product of the wavelength and the error in number of cycles. This gives a positioning error in the magnitude of centimetres. The interpolation of ground station signals gives a virtual signal, known as a Continuously Operating Reference Station (CORS) network that improves the reference signal accuracy by eliminating falsely initialised stations [15]. This component provides a Selective Availability Anti-Spoofing Module (SAASM) which guarantees secure uninterrupted GPS connection, safe-guarded from potential disruptors [16]. Receiver Autonomous Integrity Monitoring (RAIM) and Fault Detection and Exclusion (FDE) is used to eliminate error in GPS signal by comparing the signals of minimum 6 satellites and using statistical methods to identify and discard outliers from the navigation solution [17] [18].

3.7 Satellite transmitter and receiver

In order to man the UAV from a remote cockpit, a significant amount of data must be streamed between the UAV and the Ground Control Station (GCS). All flight control data and mission inputs must be streamed with minimal latency to the UAV, and vice-versa for all UAV sensor information (inertial states, positioning, video streams, lidar and HUMS). The GETSAT NanoSat H was chosen for this operation due to its small profile and weight (3.5kg) and it's high bandwidth at 4 Mbps [19]. The data stream provided operates in the K_a band portion of the microwave part of the electromagnetic spectrum between 27.5 – 30 GHz, holding more data than traditional stream in the K_u band. Furthermore, the K_a band suffers less interference from background microwave signals and is less susceptible to 'rain fade' [20]. In the case of unplanned disconnection between UAV and GCS a re-acquisition time less than 100ms will ensure there is minimal disruption of data stream.

3.8 Enhanced Ground Proximity Warning System

As mandated by UAS IFR, it is required that all unmanned aircraft have Detect and Avoidance (DAA) capability [21]. Therefore, to prepare the aircraft for future missions, namely Search and Rescue, the aircraft must be adaptable to operating in hazardous terrain. A Honeywell Mark XXII has been selected for it's recommended guidance system in the case of the helicopter entering Autorotation [22].

3.9 Miscellaneous

- Flight recorder – A Honeywell HFR5-D Flight recorder is to be placed on board, capable of a 25-hour recording duration. This component will preserve all on-board data regardless of where an unexpected landing may occur, at sea or extreme temperatures [23].

- HUMS – A Health and Usage Monitoring System is implemented to track and analyse aircraft subsystems and predict and diagnose potential failures in individual sub-systems. An in-depth review of the HUMS can be found in related report [24].
- Weather Radar – A Garmin GSX70 weather radar will provide strategic route planning in the event of unexpected weather patterns [25]. This system is optimised to advise the pilot on fuel saving routes in the case of high winds and temperature safe routes when rotor airframe icing becomes a potential hazard.
- Cooling fans – A setup of BendixKing KA-33's will be used to ensure safe operating temperatures in the avionics hub to avoid the risk of component or computer overheating [26].
- Backup battery - An EaglePicher SAR-10207 Lithium-ion battery will provide a stable power supply to the avionics in the case of loss of power production, detailed further in section 7.1

Components are to be stored away in the accessible maintenance stack detailed in Figure 2. The satellite receiver transmitter will be mounted separately and closer to the nose. A summary of all components are shown in Table 4.

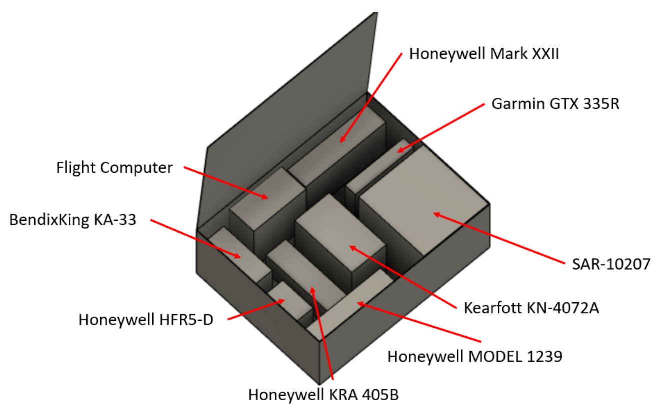


Figure 2: Avionics stack layout

Table 4: Summary of avionics components (n/a indicates that this information has not been made available)

Equipment	Manufacturer	Model	Weight (kg)	Volume (m^3)	Power (W)
AFCS	Genesys	IHR HeliSAS	16	0.002	n/a
Cooling	BendixKing	KA-33	0.55	0.0025	14
HUMS	Honeywell	MODEL 1239	1.8	0.0017	n/a
Radar altimeter	Honeywell	KRA 405B	1.4	0.0019	23.4
Black Box	Honeywell	HFR5-D	4.5	0.0034	12
Transponder	Garmin	GTX 335R	1.18	0.0015	200
INS/GPS	Kearfott	KN-4072A	5	0.0005	36
Data stream	GETSAT	NanoSat H	3.5	0.00018	70
TAWS/EGPWS	Honeywell	Mark XXII	1.8	0.0037	15
Weather radar	Garmin	GSX70	4.2	0.0052	40
Backup power supply	EaglePicher	SAR-10207	28.1	0.028	0
Total			68.03	0.0505	410.4

4 Pilot vision

As the Helicopter has been designed to be piloted remotely, the vision available to the pilot will be purely from the cameras mounted on the helicopter, therefore effort has been made to ensure the pilot feels they have the sensory input required to fly the helicopter from a remote station. When designing the optical vision to be provided to the pilot, the Endsley definition of Situational Awareness [27] was consulted:

- the perception of the elements in the environment within a volume of time and space
- the comprehension of their meaning
- the projection of their status in the near future.

In terms of optical vision, this can be quantified by:

- The total range of vision provided
- The resolution or quality of image
- Frame rate of image and perception of depth

The main source of vision is to come from an AVT CM62 gimbal mounted camera, which was chosen for it's small profile [28]. When deciding on the placement, the underbelly was first considered however this would have restricted vision to below the helicopter and increased the drag on the body. It was decided to place the gimbal camera integrated into the nose where full 180°x180° vision could be provided. A transparent hemisphere is to be placed around the camera to reduce the drag on the nose whilst preventing distortion of the image. The field of view provided by this camera can be found in Figure 3. The large range of vision provided by this camera, paired with the equipped 50x optical zoom help cover the first two points in Endsley's definition.
This vision is to be supplemented by a setup of 5 smaller 95° field of view cameras. These are to be stitched together live, with an in house produced software or off the shelf package (Imeve) [29], to produce a 360°x65° view, shown in Figure 4. This will provide the ground team with the option of cycling through camera feeds or creating a virtual cockpit, based on pilot experience [30]. This set up provides additional range of vision, beyond that of a traditional cockpit experience and aims to improve the familiarity of flying a UAV.
A belly mounted 48° Meggit CMZ is to be placed on the underbelly of the fuselage to provide a detailed view of the payload and capture mechanism Figure 5 [31]. This camera is mission specific and will be used primarily during the capture procedure.
Lastly, a 185m range Leddar Vu8 Lidar camera is to be mounted on the belly of the helicopter [32]. When flying over terrain with varying height (a forest or mountain side), this camera will provide the pilot a strong understanding of the obstacles and hazards below the helicopter. More importantly, this information will be fed to the flight computer for executing pre-programmed landing manoeuvres.

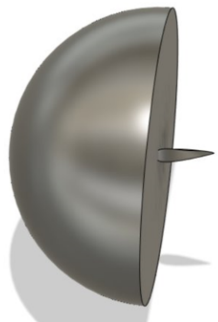


Figure 3: 180x180° vision provided by nose mounted gimbal camera

Figure 4: 360x65° vision provided by wall mounted static cameras

Figure 5: Vision provided from payload focused belly mounted camera

5 Anti-ice

At combinations of high altitudes, low temperatures and high speeds, water particles can freeze on collision with the leading edge of a rotor. The accumulation of ice can significantly effect the geometry of the airfoil, which will in turn effect the drag characteristics hence influencing the separation point of flow over the airfoil. Due to this, the rotors are to be equipped with a de-icing system to safeguard the performance of the blades. A pneumatic method was discarded due to the requirement of pressurised air through the rotor. An electro-thermal solution was then considered. Based on the power requirements ($3.9 \times 10^4\, W/m^2$) [33] and the total surface area of the heating strips, detailed in [34], a power requirement of 28 kW was obtained.

6 Actuators

The function of the actuation system is to enact the flight control inputs, the collective and cyclic (via the swashplate) and the stabilator. The helicopter will require 4 actuators per rotor, 1 on the drive shaft, 2 rotor brakes [35] and a pair on the stabilator [36]. The main design factors for picking the mechanism were:

- Force and velocity available
- Reliability and safety
- Energy efficiency

The actuation mechanisms considered were electromechanical, hydraulic and electro-hydrostatic. A summary of their advantages and disadvantages are shown in Table 5. Although unideal, the failure rate has been taken from the average failure rate in pitch, roll and yaw inputs on a fixed wing aircraft [4].
Although electromechanical actuation displays a lower failure rate, an Electrohydrostatic actuation system has been selected due to it's significant weight savings and energy efficiency.

Table 5: *Summary of actuation types*

Actuation type	Advantages	Disadvantages	Average failure rate (1/h)
Electro-mechanical	High efficiency Easy installation and maintenance	Risk of gear jamming Backlash	5×10^{-6}
Hydraulic	High forces High reliability	Low efficiency Requires pipe system Risk of leakage	5.3×10^{-5}
Electro-hydrostatic	High efficiency High power density	Less commonly used	9.3×10^{-5}

6.1 Electro-hydrostatic actuation

The EHA system is the next generation of 'fly-by-wire' systems, known as 'power-by-wire' used in modern aircraft where weight saving is a priority, such as the F-35 Lighting II series. The main difference between an EHA system and hydraulics is that the EHA system is self-contained and placed locally at the control surface. Each unit contains their own pump, accumulator and hydraulic fluid. By removing the global fluid reservoir, significant weight savings are made by reducing the amount of on-board fluid to the exact amount required. Similarly, the steel tube fluid distribution system is removed which also significantly reduces total wall friction. This makes the system more energy efficient and reduces the required cooling on the fluid. Eliminating the reservoir also removes the need to continuously run the main pump in the reservoir, thus reducing the electrical power demand. A schematic of the operation of an EHA system is shown in Figure 6

In an EHA system, a control input is converted to a servo action which drives the variable speed fixed displacement pump. Rather than controlling the flow of the fluid using a set of control valves, the flow is directed by the velocity of the pump. The overall result is that the pressure in the chambers are solely dependent on the hydraulic power demand.

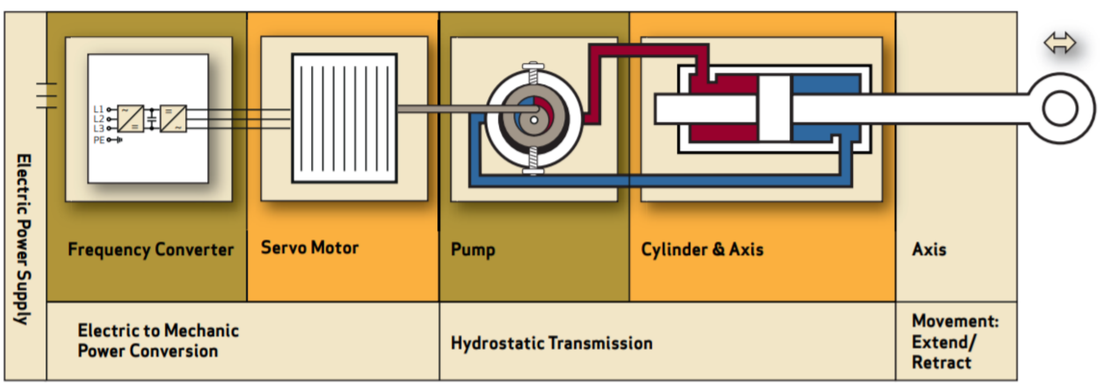


Figure 6: Schematic of Electro-hydrostatic actuator [2]

6.2 Hydraulic fluid

When choosing a hydraulic fluid, considerations must be made for the operating temperatures of the fluid and it's viscous properties. The fluid must flow easily at very low temperatures but also retain it's viscosity at high temperatures. A preferable fluid would have a very low freezing point and a very high boiling point. It also needs to have lubricating properties and be anti-corrosive. Figure 7 shows the variation of viscosity with temperature for multiple hydraulic fluids. It can be seen that MIL H5606 shows the least variation, therefore will be used in the EHA system.

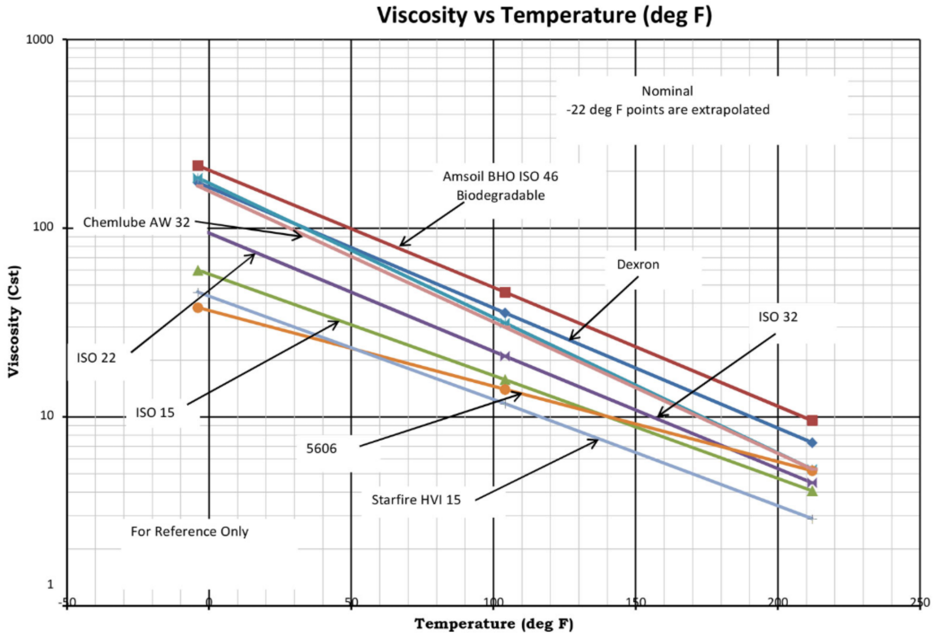


Figure 7: Variance of fluid viscosity with temperature of multiple hydraulic fluids [3]

6.3 Actuator failsafe

It is standard practice for flight control mechanisms to have triple redundancy, making the chance of loss of control of all EHA systems negligible. However, in the event of total flight control loss, the EHA still has integrated fail-safe methods. The addition of an extra accumulator and chamber (C) in the final EHA unit would allow for a full deflection that can be locked in, shown in Figure 8. In practice, the actuators will be deflected such that the rotors are fixed at +4° as this pitch causes no blades to stall and produces the most lift [37].

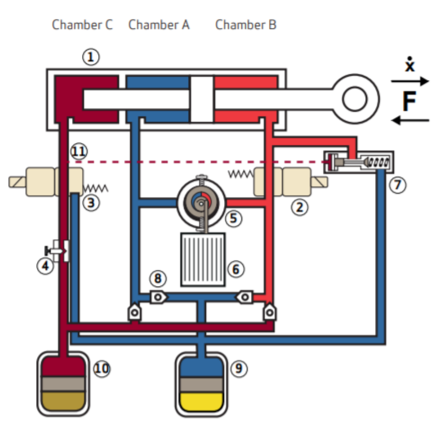


Figure 8: Electro-hydrostatic actuator equipped with a reserve accumulator [2]

6.4 Actuator sizing

When sizing the actuators, the first consideration made was the deflection of actuators. The total controllable range of the main rotors are ±12° and knowing that the actuators should be capable of a full deflection in one rotation at 488 RPM allows for the maximum angluar velocity to be calculated, 3.4 rad/s. The maximum power requirement could then be found using Equation 1

$$P = \frac{M\dot{\theta}}{\eta} \tag{1}$$

$$M = 0.5\rho C_M c^2 \int (V_x + \omega r)^2 dr \tag{2}$$

Where M is the maximum pitching moment generated by a blade and the EHA efficiency $\eta = 0.8$ [4]. The maximum pitching moment was calculated using Equation 2 where $C_m = 0.009$ for the airfoil used [38]. After a 1.5x safety factor was applied, the power required by each actuator was found to be 1337 W. The pitch horn was taken to be 85mm, the same as on a UH-60 [39]. Table 6 shows the actuator parameters. Lastly the weight of each actuator was estimated from historical data to be 11.1kg, shown in Figure 9.

Table 6: Dimensions of Actuator

Parameter	Value
Pitch horn length (mm)	85
Stroke length (mm)	35
Maximum velocity (m/s)	0.29
Maximum load (N)	2531
Weight (kg)	11.1

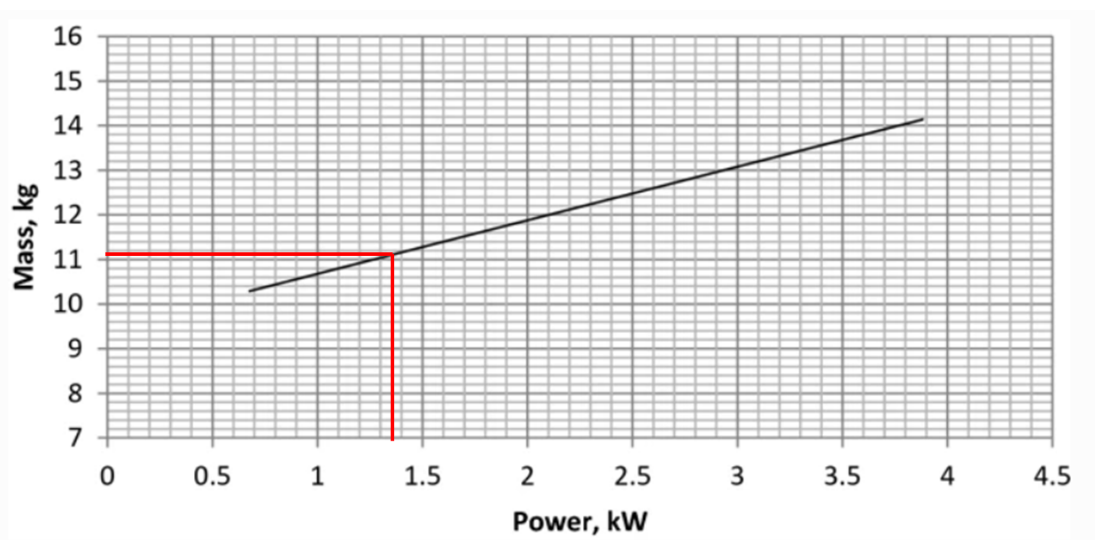


Figure 9: Linear regression relationship between mass and power of 9 in use electro-hydrostatic actuators [4]

7 Power generation

Table 7 shows a breakdown of the total on board power requirements.

Table 7: Summary of power requirements

Component	Power (kW)
Anti-icing	28.0
Rotor actuators	14.7
Avionics	0.50
Stabilator actuators	0.45
Total	43.65

Based on a 45kW power requirement, available starter generators on the market were shortlisted to those in Table 8. The Emrax 188 was selected due to having the highest power density. In order to produce a maximum of 45 kW, a 5000RPM input is required, shown in Figure 10. Power generation is 95% efficient at this RPM [5]. A 1.28:1 reduction gear will be placed on the generator to feed power from the 6400RPM stage in the transmission [35].

Table 8: Summary of shortlisted generators

Generator	Energy density (kW/kg)	Weight (kg)
Honeywell 45 kVA	3.5	12.8
Siemens 65 kVA	5	13
Emrax 188	7.22	7.2

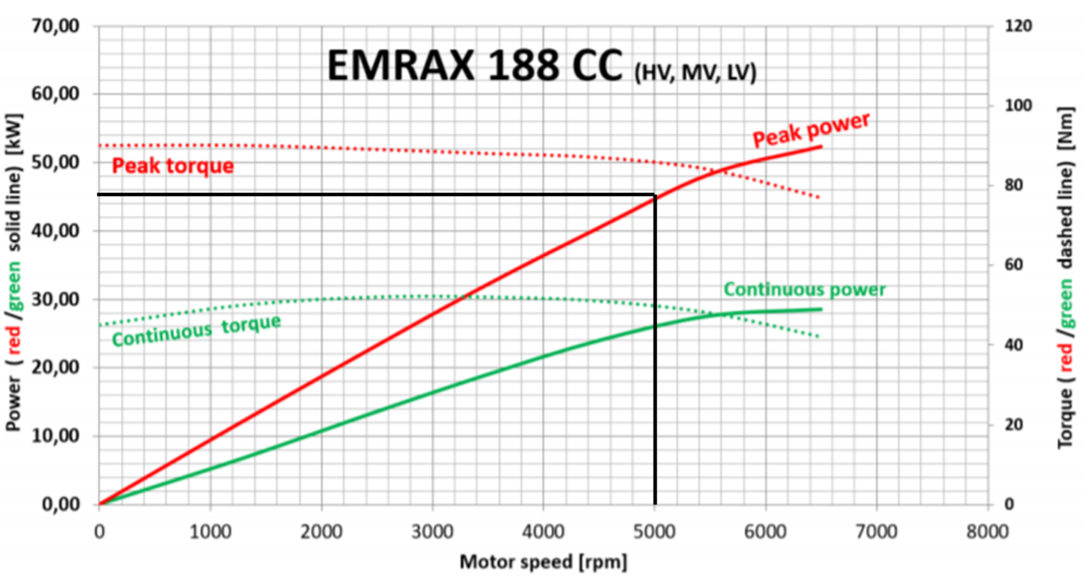


Figure 10: Power curve of Emrax 188 [5]

7.1 Battery power

As most of the chosen avionics units require 28VDC, the power directed to the avionics will first pass through a converter to produce Direct Current, followed by a transformer to fix the voltage at 28VDC. A 28 VDC battery is to be placed in the avionics bay, this battery is required for providing redundant electrical power to the avionics in the case of generator failure. A lithium ion battery is to be used as these have the highest power density (0.34kW/kg) [40]. These batteries are also considered safer than other common types of batteries, like lead acid batteries which can begin to release hydrogen gas after operating at high temperatures. An EaglePicher SAR-10207 battery was chosen [41]. In the case of power plant compressor surge, this battery has the storage capacity to provide the energy required to achieve minimum RPM (1.7MJ), calculation available in Appendix A. In the event of engine, transmission or generator failure, the rotor clutch will disengage and force the helicopter into auto-rotation. The backup battery can fully power the avionics and provide limited power (1.8kW) to the actuators for 80 minutes. This should give the pilot enough time to find a safe landing zone.

8 Conclusion and Further Iterations

Based on feedback from the simulation team [42], more work can be done to reduce the pilot workload when completing the capture manoeuvre. The field of view of the belly camera was not large enough to have a consistent view of the payload. Similarly, banking of the helicopter shifted the orientation of the camera and lost sight of the payload. This could be solved by using a downwards-stabilising camera that will constantly face the ground and subsequently the payload. Another potential solution would be to mount a second gimbal mounted camera on the rear. This may call for the input of the second pilot, to manage the rear camera and rudder grapple whilst the first continues to fly the helicopter. Data gained from the lidar camera should be manipulated to better inform the pilot of the distance between the grapple mechanism and the payload.

Currently the largest draw on power is the anti-icing system at 28 kW. In a later iteration an ultrasonic frequency distortion method should be considered. This entails a series of actuators, of millimetre thickness, along the leading edge. These can vibrate at 28.5 KHz and produce shear stresses capable of delaminating ice from the blade. This method is shown to have a power consumption of $1.9 \times 10^3\ W/m^2$ [33], corresponding to a 95% decrease in power consumption compared to an electro-thermal method. This system can be integrated into the blade during the next iteration of blade structural design.

More accurate methods for sizing the EHA power requirement and the engine re-start energy should be considered. Whereas in this report Newtonian mechanics equations were used, an approach requiring larger involvement with the flight control and powerplant team should be considered.

The actuation method can be further improved upon by rethinking the swashplate design and introducing individual blade control which is not only more efficient for the actuation system but also improves overall rotor performance and assists in reducing rotor vibration, hence reducing the maintenance required [43]. Currently, each actuator has full power triple redundancy, weight saving can be made by considering 75% power triple redundancy. If an actuator fails in this system, there are still 2 available actuators than can provide 150% of the maximum power required. This can be done in a detailed design of the actuators.

This project has proven a feasible layout of sub-systems. As recognised in this report, when the helicopter is moved to a fully autonomous platform a reassessment of the avionics must be made. More focus must be put on the flight computer algorithms and the sensing units should be thoroughly scrutinised to ensure they can provide the required information for perfect decision making.

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A Appendicies

A.1 Rotational energy of powerplant

Dimensions of the powerplant stages are detailed in Table 9, note that all compressor stages have been combined. These values were used to find the moment of inertia of the rotational elements of the powerplant using equation 3. The Engine rotational energy was calculated using Equation 4, knowing that the minimum operating RPM is 18000. Required rotational energy is 1.68 MJ.

$$I = \frac{1}{2}mr^2 \tag{3}$$

$$E_k = \frac{1}{2}I\omega^2 \tag{4}$$

Table 9: Dimensions of powerplant components

Component	Mass (kg)	Radius (m)
Compressor stages	33	0.152
High pressure turbine	50	0.15